

MERIT: Retrieval Is Not Contribution — Counterfactual Credit Assignment for Self-Evolving Agent Memory

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Abstract

Self-evolving language-model agents distill interaction trajectories into a persistent memory bank, and the quality of that self-improvement loop depends on how credit is assigned to individual memories. We argue that the credit signals used by current systems are *correlational*: a memory that merely co-occurs with success is reinforced, so *retrieval is mistaken for contribution*. We hypothesize that this reinforces *superstitious memories*—entries with zero or negative causal effect that nonetheless accumulate utility and occupy retrieval slots—and that their fraction grows with deployment length. The difficulty is that a memory’s causal value is defined only counterfactually, by how much outcomes degrade when it is removed, and per-memory intervention appears unaffordable in an online, non-stationary, closed loop. We propose MERIT, which is designed to make counterfactual credit affordable through a strictly linear pipeline: Randomized Interventional Trials (RIT) collect small-budget causal labels, Amortized Counterfactual Attribution (ACA) compresses them into an $O(1)$ -per-event score, and two minimal consumers—threshold governance and scope-gated retrieval—read the same signal. We plan a systematic evaluation testing whether ACA recovers causal credit at low cost, whether its consumers flatten superstition and reduce cross-task interference, and whether MERIT improves average success over the strongest baseline; the planned headline gain is [TBD], pending verification.

Introduction

Language-model agents are moving from stateless responders to experience-driven systems that improve over time by distilling their own interaction trajectories into a persistent memory bank and reusing it through a retrieve–execute–write loop [TBD]. The implicit premise of this paradigm is that the system knows *which memory deserves credit*: writing, merging, evicting, and retrieving all depend on a per-memory utility signal. Yet the signals in widely used systems are, on the characterization we adopt from prior descriptions and subject to bibliographic verification, *correlational*. When a task succeeds and a memory was retrieved, that memory’s utility is incremented. This conflates three distinct events:

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a memory can be *retrieved* without being *used*, and used without providing genuine *causal* help.

Retrieval is not contribution. A memory’s value is defined only counterfactually—by how much task outcomes degrade when it is removed from the retrieved set. Correlational credit does not measure this quantity. We hypothesize that it behaves like the reinforcement schedule in Skinner’s superstition experiment [TBD]: memories that merely co-occur with success are persistently strengthened, producing *superstitious memories*—entries with zero or negative causal contribution that nonetheless rise into the top-utility set and occupy retrieval slots. Because retrieval probability tends to track query solvability and high utility feeds back into higher retrieval rank, we hypothesize a self-reinforcing loop in which the fraction of superstitious memories *grows* with deployment length. A second, related failure is cross-task interference: a memory that is genuinely useful for one task family is indiscriminately retrieved into another and causes harm.

Why prior signals are insufficient. Existing approaches in our comparison commonly use observational, co-occurrence credit: success-trajectory distillation [TBD], environment-feedback utility [TBD], and Monte-Carlo or reinforcement updates over outcomes [TBD]. What is learned is the management *policy*; the underlying signal remains correlational. We distinguish passive co-occurrence counting from outcome-reward reinforcement learning under exploration, which is correlational only in a weaker sense; the diagnostic below is scoped to the systems and streams we evaluate, not to all learned managers. Counterfactual data-valuation methods such as Data Shapley [TBD], influence functions [TBD], and TracIn [TBD] target static, offline training data, while context-attribution methods [TBD] explain a single generation. None, to the best of our current and still-to-be-verified survey, deliver counterfactual credit in an online, non-stationary, closed loop where attribution itself changes the data distribution.

Approach. We propose MERIT, which replaces correlational bookkeeping with counterfactual credit while keeping it affordable. MERIT adds three tightly coupled components to the standard loop, in a strictly linear dependency $\text{RIT} \rightarrow \text{ACA} \rightarrow \text{consumers}$: (i) *Randomized Interventional Trials* (RIT), a small-budget paired leave-one-out protocol

that produces interventional outcome-difference labels; (ii) *Amortized Counterfactual Attribution* (ACA), an $O(1)$ -per-event attributor trained on those labels and periodically recalibrated to track closed-loop drift; and (iii) two deliberately simple consumers of the resulting score—threshold-based credit governance and scope-gated retrieval. Figure 1 contrasts correlational credit with this pipeline. Our central position is that *causal credit—not better heuristics—is the missing primitive of self-evolution*.

Contributions.

- **(C1) Superstitious memories.** We formalize and plan to quantify a diagnostic phenomenon in the memory systems and streams we evaluate: correlational credit breeds zero/negative-contribution memories in the top-utility set whose fraction rises with deployment time, and we introduce an interventional audit (Rrr) with two statistics—the credit–contribution correlation (CCC) and the superstition rate ($SR@k$)—to measure it.
- **(C2) Amortized counterfactual attribution.** We introduce ACA, which amortizes interventional labels into an $O(1)$ -per-event contribution score with periodic recalibration, and we plan to test whether it recovers causal credit substantially better than the correlational baseline at low cost.
- **(C3) Directly consumable causal credit.** We contribute a design in which a single causal-credit signal is directly consumable: threshold governance and scope-gated retrieval are two readouts of the same score. Governance, scope gating, and a per-operation reward substitution are evaluated as consumers of this one signal, not as separate methods, and we plan to test whether they respectively flatten superstition accumulation and reduce cross-task interference.
- **(C4) Systematic evaluation.** We plan a systematic evaluation across four benchmark categories, three backbones, and three seeds, testing whether MERIT improves average success over the strongest baseline at low token overhead.

Planned evidence. Every empirical statement in this paper is a planned test, not a measured result; unresolved quantities are written as registered ledger placeholders. A diagnostic pilot (Section 4) tests whether superstition accumulates; the mechanism analysis (Section 6) tests whether ACA recovers the audited signal in closed loop under the same axes as the pilot; and the main study tests whether the recovered signal translates into a planned average gain of [TBD] over the strongest baseline. Each claim carries an explicit falsification gate, and results that fail a gate will narrow the corresponding claim rather than be suppressed.

Related Work

We organize prior work by the credit signal it uses, not paper by paper, and position MERIT against each family. All bibliographic metadata is pending verification, and we avoid unqualified priority claims.

Self-evolving memory and experience distillation. A first family equips agents with a persistent memory that is written and reused across tasks: general memory stores [TBD], agentic memory with linked notes [TBD], workflow memory [TBD], reasoning-trajectory banks [TBD], skill or exemplar memory [TBD], and evolving stores [TBD]. On the characterization we adopt from their descriptions (pending verification), the credit these systems attach to a memory is heuristic or co-occurrence based: a retrieved memory inherits the outcome of the task it accompanied. MERIT differs by auditing the causal validity of that credit with interventional evidence, rather than proposing another heuristic for accumulating it.

Learned memory management. A second family learns the memory-management policy with reinforcement or Monte-Carlo signals: reward-driven managers [TBD], policy-optimized memory agents [TBD], value-learned stores [TBD], and skill-management policies [TBD]. What is learned here is *which action to take*, while the reward that drives learning remains an outcome-level, correlational quantity. Complementary work improves write-time quality through multi-agent consensus [TBD], which is orthogonal to and composable with fixing full-lifecycle credit. MERIT targets the signal rather than the policy: it replaces the reward itself with a counterfactual score, and we later test whether substituting that score into such a manager lifts its performance.

Data valuation and context attribution. A third family measures the influence of data or context on model behavior: Data Shapley [TBD], influence functions [TBD], and TracIn [TBD] value static training points offline, while context attribution [TBD] explains a single generation. These methods share MERIT’s counterfactual framing but operate in a static, offline, or single-shot regime. The setting we address is online, non-stationary, and closed-loop: the attribution signal changes the retrieval distribution, which in turn changes future labels. This is what motivates amortization and periodic recalibration in ACA.

Interference and orthogonal directions. Cross-task interference in mixed-stream deployment has been observed [TBD] [TBD], and test-time reasoning methods [TBD] [TBD] improve behavior along an axis orthogonal to memory credit; we treat the latter as compatible baselines. The capability comparison in Table 5 summarizes where MERIT is intended to sit relative to these families; every competitor cell remains unresolved until its cited source is checked item by item, and an unresolved cell must not be read as a lack of capability. The table appears with the experiments (Table 5).

Problem Formulation and Superstitious-Memory Hypothesis

Setting and notation. An agent processes a stream of tasks $\{(q_t, r_t)\}$, where q_t is a query or task and $r_t \in [0, 1]$ is an outcome signal (an environment success indicator or a consistency-checked judge score). It maintains a memory bank $\mathcal{M}_t = \{m_i\}$, a retrieval operator $R(q_t) = C_t \subseteq \mathcal{M}_t$

Fig. 1 placeholder: Problem and solution schematic

Left panel: correlational credit and a top-utility memory set, with memories of non-positive causal contribution highlighted.
 Right panel: the $R_{IT} \rightarrow ACA \rightarrow$ consumers pipeline on the retrieve-execute-write loop.
 Future source: pilot SR@k and mechanism analysis. No result geometry is drawn at this stage.

Figure 1: Planned hero schematic contrasting correlational credit with the counterfactual-credit pipeline. The eventual empirical annotation remains TBD until verification.

returning a top- k set, and a policy $\pi(y | q, C)$ that produces an output y . **MERIT** adds an attributor g_θ , a per-memory running credit $\hat{\varphi}_i$, and a scope representation S_i . These are the core symbols; Algorithm 1 lists the full runtime state (label pools, trajectories, per-memory counts, prototypes, quarantine and calibration state).

Immediate retrieval contribution. We define a memory’s value by intervention, at a single retrieval event and conditioned on the deployment history.

Definition 1 (Immediate retrieval contribution) *For a retrieval event with query q , realized retrieved set $C = R(q)$, and history H_t ,*

$$\varphi_{i,t}(q, C) \triangleq \mathbb{E}_\xi[r_t(C; \xi) - r_t(C \setminus \{m_i\}; \xi) | H_t], \quad (1)$$

the expected outcome difference from removing m_i , over policy and environment randomness ξ .

This is the single-event effect, *not* a long-horizon memory value; long-horizon value is out of scope. Higher-order interactions within C are handled by AS3 and the group-intervention variant (A7), not by φ . We separate four related quantities and never conflate them: the true effect φ ; its finite-rollout R_{IT} estimate $\tilde{\varphi}$ (Definition 1 evaluated with K paired rollouts); the amortized ACA prediction $\hat{\varphi}$; and the online running aggregate $\bar{\varphi}_i$ (an EMA of $\hat{\varphi}$). At the memory level we distinguish the true aggregate target $\Phi_i \triangleq \mathbb{E}[\varphi_{i,t}]$ over the audit distribution from its finite R_{IT} audit estimate $\hat{\Phi}_i$ (reported with a confidence interval; Section 4). The baseline signal, by contrast, is associational.

Definition 2 (Co-occurrence utility) *The baseline memory-level signal is*

$$\hat{U}(m_i) = \frac{\sum_t \mathbf{1}[m_i \in R(q_t)] r_t}{\sum_t \mathbf{1}[m_i \in R(q_t)]}, \quad (2)$$

the average outcome of tasks in which m_i was retrieved.

Why correlational credit can be superstitious. Because φ is a definitional removal difference, the memory-level decomposition

$$\mathbb{E}[r(C) | i \text{ retr.}] = \underbrace{\mathbb{E}[r(C \setminus \{m_i\}) | i \text{ retr.}]}_{\text{leave-one-out baseline}} + \underbrace{\mathbb{E}[\varphi_i | i \text{ retr.}]}_{\text{mean contribution}} \quad (3)$$

is an *identity*. Hence $\hat{U}_i$ mixes the leave-one-out baseline success with the mean contribution and, whenever that baseline term is nonzero, is not a pure estimator of contribution. We state this as a hypothesis, not a theorem.

Conjecture 1 (Superstition bias, informal) *$\hat{U}_i$ differs from $\mathbb{E}[\varphi_i | i \text{ retrieved}]$ by exactly the leave-one-out baseline term; whenever that term is nonzero, $\hat{U}_i$ is a biased estimator of contribution. If retrieval is statistically dependent on baseline solvability, this dependence amplifies and systematizes the bias, and under positive feedback in which utility participates in retrieval ranking a stable superstition equilibrium can arise, with $\hat{U}_i$ high while $\varphi_i = 0$.*

Retrieval-solvability dependence is a mechanism that may amplify the bias under the conditions tested in the pilot; it is *not* necessary for the bias to exist or for the decomposition identity to hold. A formal statement and proof are deferred; Conjecture 1 is not proven in this paper and must not be read as established.

Assumptions. Three assumptions scope the method and are stated for rebuttal. To avoid collision with the ablation labels A1–A7 used later, we name the assumptions AS1–AS3.

- **AS1 (observable outcome).** Each task admits a usable r_t , either an environment success signal or a consistency-checked judge score.
- **AS2 (local replayability).** A task subset can be replayed from a fixed initial state (e.g., container snapshots or text environments). Only R_{IT} label collection requires AS2; ACA online scoring does not.
- **AS3 (low-order interaction).** Interaction effects within a retrieved set are bounded, so leave-one-out error against the Shapley value is controllable. Cases that violate AS3 (redundant memory pairs) are surfaced in the boundary study with a group-intervention fallback.

Objectives. ACA learns g_θ by minimizing a regression loss on R_{IT} labels, $\mathbb{E}[(g_\theta(x_i(q)) - \tilde{\varphi}_i(q))^2]$, where $x_i(q)$ is a feature vector for the event. The system objective maximizes streaming cumulative success $\sum_t r_t$ subject to a token budget: retrieval plus R_{IT} sampling stay within $1 + \varepsilon$ times the baseline cost, with $\varepsilon = [TBD]$ as a provisional hyperparameter.

Pilot and Diagnostic Study

This section diagnoses the failure that MERIT is designed to address, on the specific memory systems and task streams we evaluate; it introduces no second method and makes no claim about self-evolving memory in general. Every plotted quantity is an unresolved pilot output, and the study is governed by a prespecified falsification gate.

Systems and streams. We reproduce two credit designs on a unified codebase—a heuristic-credit bank in the style of ReasoningBank [TBD] and a Monte-Carlo-learned bank in the style of MemRL [TBD]—using the same backbone and retriever. We run them on two replayable streams (a text embodied environment and a multi-hop retrieval task), which are required because interventions must re-run paired rollouts from an identical initial state (AS2). Stream length and seed count are fixed as a planned configuration.

Two-stage interventional protocol. This is the first appearance of RIT. Thresholds of scientific meaning are fixed *before* the pilot: the equivalence half-width δ (the smallest memory-level contribution we treat as practically meaningful) and the minimal effect of interest are pre-registered as user-approved values, not read from pilot noise. Stage 1 is a pilot with $K = 5$ paired rollouts that estimates variance; Stage 2 then computes and freezes the formal K , the per-memory audit sample size n^{audit} , and the seed count so the confidence-interval half-width falls below the pre-fixed δ . The confirmatory statistics use data disjoint from the Stage-1 pilot.

At snapshot points $t \in \{100, 200, 300, 400, 500\}$ we sample retrieval events and, for each audited memory, perform a paired leave-one-out comparison—the full retrieved set versus the same set with the memory removed—holding context order fixed and inserting a length-matched neutral pad whose neutrality is validated separately ([TBD]). The event label is $\tilde{\varphi}_i(q)$, the K -rollout success-rate difference. To keep audit estimates unbiased, the audit pool is a random or stratified probability sample with known inclusion probabilities (any priority sampling is confined to the training pool), and estimates are restricted to memories with adequate common support (coverage [TBD]). Rollout counts and snapshot points are a planned configuration; the amortized cost is [TBD] per event, and the totals [TBD] and [TBD] appear in an appendix cost table.

Statistics. We separate the true memory-level aggregate target $\Phi_i = \mathbb{E}[\varphi_{i,t}]$ from its finite RIT audit estimate $\tilde{\Phi}_i$, reported with a task-clustered paired confidence interval; $\tilde{\Phi}_i$ is an estimate, not ground truth. The *credit-contribution correlation* at the memory level, $\text{CCC}(s, \tilde{\Phi})$ for a ranking signal $s \in \{\tilde{U}, \tilde{\varphi}\}$, is the Spearman correlation between the signal and the audit estimate. The *superstition rate* $\text{SR}@k(s)$ is the fraction of the top- $k\%$ memories by s whose contribution is non-positive, judged from the CI of $\tilde{\Phi}_i$ against the pre-fixed equivalence band $[-\delta, \delta]$ —not a bare ≤ 0 cut. We report $\text{SR}@20\%$ split into a *harmful* subset ($\tilde{\Phi}_i$ confidently below $-\delta$) and a *practically null* subset (CI within $[-\delta, \delta]$); redundant-but-useful memories are separated and treated in

Fig. 2 placeholder: Baseline diagnostic

(a) x : system utility; y : counterfactual contribution ϕ .

(b) x : task index t (100–500); y : $\text{SR}@20\%$.

Methods: ReasoningBank and MemRL.

The axis ranges and statistical definitions are locked to Fig. 5.

Figure 2: Planned diagnostic of credit-contribution correlation and superstition rate over deployment. All plotted values remain unresolved pilot outputs.

the AS3 boundary study. Figure 2 plots the baseline memory-level CCC and the $\text{SR}@20\%$ trajectory over deployment.

Planned readings and controls. We test whether the baseline memory-level correlation is low ([TBD] for the heuristic bank and [TBD] for the learned bank) and whether the superstition subsets rise over deployment: harmful from [TBD] to [TBD] and practically null from [TBD] to [TBD], with the combined $\text{SR}@20\%$ moving from [TBD] to [TBD] (each a point estimate with its CI), as predicted if correlational credit does not self-correct. Two controls guard against alternative explanations. A split-half re-test of $\tilde{\varphi}$ gives reliability [TBD], testing whether a low correlation is a measurement artifact. Removing the entire positive-contribution subset and measuring the success drop [TBD] tests whether memory has value at all, so that a defective credit assignment—rather than useless memory—is the operative problem.

Falsification gate. If the memory-level credit-contribution correlation exceeds 0.5, or if the superstition subsets do not rise with deployment, the attribution hypothesis is not supported; we will then narrow the claim to heterogeneous streams rather than force a positive reading. The diagnostic yields two requirements for the method: approximate the RIT credit online, and expose the same credit signal to downstream consumers. Both are realized by MERIT, whose architecture is shown in Figure 3.

Method: MERIT

ACA amortizes the RIT estimate $\tilde{\varphi}$ of the same target φ audited in Figure 2, producing a per-event prediction $\hat{\varphi}$; it introduces no new quantity. MERIT adds three components to the standard retrieve-execute-write loop in an architecturally linear dependency, $\text{RIT} \rightarrow \text{ACA} \rightarrow \text{consumers}$ (Figure 3). All thresholds and coefficients below are provisional hyperparameters.

Randomized Interventional Trials

RIT continuously produces interventional outcome-difference labels at low budget. With sampling probability $p = [\text{TBD}]$ (a provisional hyperparameter; the source default is 5%), a replayable task enters the trial group. For each audited memory, RIT performs the paired leave-one-out comparison of Section 4 with the frozen formal K (Stage-1 pilot

Fig. 3 placeholder: MERIT architecture

Retrieve-execute-write loop with three ordered additions: paired-LOO RIT labels, amortized ACA scores with recalibration, and two consumers (credit governance and scope-gated retrieval).
This is a data-free schematic and can be drawn before experiments finish.

Figure 3: Planned architecture of the strictly ordered RIT $\rightarrow$ ACA $\rightarrow$ consumers pipeline.

$K = 5$) and the validated neutral pad, and emits a quadruple $(q, m_i, \text{trajectory}, \tilde{\varphi}_i)$ into a label pool split into disjoint training and audit parts. Sampling runs only on tasks that satisfy AS2. Because the naive alternative—letting the agent self-judge whether a memory helped—can reproduce a self-confirmation trap, we retain it only as the ablation variant A-judge rather than as the label source.

Amortized Counterfactual Attribution

ACA compresses the training labels into a per-event prediction $\hat{\varphi}$, scored in $O(d)$ per retrieved memory at a fixed feature dimension d . The feature vector $x_i(q)$ has three groups: a query representation, a memory representation (a frozen embedding encoder), and *usage-behavior* features that distinguish a memory that was retrieved but ignored from one that was actually used— n -gram and edit overlap between memory content and output, explicit citation markers, context position and retrieval rank, and the log-likelihood gain on memory tokens. A two-layer perceptron regresses $\hat{\varphi} \in [-1, 1]$ under a Huber loss ($\delta_H = 0.1$, provisional; distinct from the SR equivalence band δ and the governance dead-zone). Because the consumers change the closed-loop retrieval distribution, ACA performs *periodic recalibration* every 100 tasks (planned configuration), fine-tuning on the newest training labels and applying isotonic calibration. Event-level fidelity is measured by CCC($\hat{\varphi}_{i,t}, \tilde{\varphi}_{i,t}$) on the held-out audit split. At inference, ACA scores each of the top- k retrieved memories and updates an exponential moving average of the running credit $\bar{\varphi}_i$ and the positive/negative query prototype sets. The trainable parameter count is $[TBD]$. Where a backbone exposes no token log-probabilities (the closed-source row), the log-likelihood feature is masked and the remaining features are used.

Two consumers of one signal

Governance and scope gating are two readouts of the same $\bar{\varphi}$; neither is an independent method.

Credit governance. Three threshold rules update retention. A memory is *evicted* when $\bar{\varphi}_i < -0.02$ and its sample count $n_i \geq 8$; two memories are *merged* when their embedding similarity exceeds 0.9 and both have $\bar{\varphi} > 0$, with the merged content inherited unchanged from the baseline writer and only the merge *trigger* credit-based (write-time content quality is out of scope); and new memories are *quarantined*, receiving a UCB-style forced exploration bonus over

their first $n_{\min} = 3$ retrievals that does not enter the ranking score, to avoid cold-start death. The ± 0.02 governance dead-zone is a governance hyperparameter only (not the SR equivalence band δ), deliberate because the sign of the estimate is least stable near $\varphi \approx 0$. Ablation A2 uses the identical ACA architecture, features, update rule and compute budget but swaps the interventional label for a matched observational/outcome label, isolating interventional supervision at matched capacity.

Scope-gated retrieval. Each memory maintains positive and negative query prototypes (the mean embedding of queries with $\hat{\varphi} > \tau^+$ and $\hat{\varphi} < \tau^-$). The retrieval score combines semantic relevance, scope agreement, and credit,

$$\text{score} = \alpha \cdot \text{rel} + \beta (\text{sim}(q, \text{proto}^+) - \text{sim}(q, \text{proto}^-)) + \gamma \cdot \bar{\varphi}_i, \quad (4)$$

with $(\alpha, \beta, \gamma) = (1.0, 0.5, 0.3)$ as provisional defaults. This is intended to address cross-task interference by learning *where* each memory applies. The same $\hat{\varphi}$ can also replace a manager’s outcome reward, which we evaluate later as a signal-portability check rather than a separate contribution.

Cost and complexity

The dependency RIT $\rightarrow$ ACA $\rightarrow$ consumers is an architecturally linear chain. ACA inference is $O(kd)$ per query (top- k memories at feature dimension d), i.e. $O(d)$ per retrieved-memory event and independent of bank size; RIT label collection and periodic recalibration are amortized over the stream and accounted for separately. Constant-per-event scoring is a design property; the end-to-end token and memory overhead— $[TBD]$ tokens and $[TBD]$ of resident VRAM—the per-event latency $[TBD]$, and the amortized RIT cost $[TBD]$ remain planned measurements. Algorithm 1 summarizes the loop.

Experiments and Analysis

This section specifies protocols, table skeletons, figure placeholders, and conditional interpretation plans. No ranking or numerical conclusion is asserted before verification; every result cell is an unresolved ledger placeholder, and each analysis states the test it will run and the reading that would support or fail its hypothesis. Table 4 states, for each claim, its atomic falsification gate: estimand, comparator, audit population, decision rule, and the claim-narrowing action on failure.

Algorithm 1 MERIT main loop

Input: stream $\{q_t\}$, bank $\mathcal{M}$, retriever R , policy π , attributor g_θ , budget p
State: training/audit label pools, running credit $\bar{\varphi}$, prototypes $\text{proto}^\pm$, counts n , quarantine set, calibration state
Init: empty pools; g_θ warm-started; new memories quarantined

```
1: for  $t = 1, 2, \dots$  do
2:    $C \leftarrow R(q_t)$  {scope-gated score; quarantine bonus not in ranking}
3:    $y_t, \text{traj} \leftarrow \pi(q_t, C)$ ;  $r_t \leftarrow \text{env/judge}(y_t)$ 
4:   for  $m_i \in C$  do
5:      $\hat{\varphi}_i \leftarrow g_\theta(x_i(q_t, \text{traj}))$ ; update  $\bar{\varphi}_i, \text{proto}_i^\pm, n_i$ 
6:   end for
7:   if  $\text{replayable}(q_t)$  and  $\text{rand}() < p$  then
8:     for  $m_i$  in sampled subset of  $C$  do
9:        $\bar{\varphi} \leftarrow \text{paired-LOO}(q_t, C, m_i, K)$ ; add to train/audit pool
10:    end for
11:  end if
12:  governance( $\mathcal{M}$ ): evict / merge / quarantine by rules on  $(\bar{\varphi}, n)$ 
13:  if  $t \bmod 100 = 0$  then
14:    recalibrate  $g_\theta$  on training pool (Huber + isotonic)
15:  end if
16: end for
```

Setup

We plan to evaluate three backbones—a primary open model, a larger open model as a scale-trend point, and a closed model as a generalization row—on a unified memory-system codebase for baseline fairness. ACA uses a frozen embedding encoder with a two-layer head. Evaluation spans four benchmark categories: streaming task streams (dialogue and agentic), a web-agent benchmark, a long-term memory benchmark, and a conversational memory benchmark; the pilot uses two replayable environments. For fairness, all methods share the backbone, retriever, and top- $k = 4$ budget, with baselines given an equal token budget for their own mechanism. We report a two-stage macro-average—streams aggregated to a benchmark-category score, then equal weight across the four categories—with task-clustered paired BCa intervals and a simultaneous max-baseline comparison under Holm correction; seeds default to $\{13, 42, 2026\}$, with the final count fixed by a pre-registered power analysis. A subset uses a cross-source judge, of which 200 examples are human-checked for agreement $[TBD]$. A single main run costs $[TBD]$, with the full cost table in the appendix.

Main results

Table 1 orders methods by lineage—no-memory, heuristic credit, learned credit, orthogonal reasoning, and an intervention reference—and orders the stream columns S1–S5 by increasing heterogeneity; the identities of S2–S4 are not specified in the source. We test three predictions against the eventual numbers: whether the gain grows with stream heterogeneity (from $[TBD]$ on the single-domain stream to

Fig. 4 placeholder: Streaming and efficiency

Main panel: x = task index; y = cumulative success. Compare MERIT, strongest verified baselines, and the RIT-Full intervention reference.

Optional inset: planned RIT-budget p sweep, shown only after data verification.

Figure 4: Planned test of whether performance differences concentrate in the later portion of the stream and whether the RIT budget reaches a Pareto-efficient region.

$[TBD]$ on the mixed stream); whether learned-credit baselines are reliably better than heuristic ones; and whether MERIT reaches $[TBD]$ of the RIT-Full reference gain at a fraction of its cost. The headline average gain over the strongest baseline is $[TBD]$. The RIT-Full row is an intervention reference and is treated as a ceiling only for quantities where that status follows by construction.

Figure 4 gives the streaming cumulative-success view of these results, with the planned RIT-budget sweep as an inset (detailed under Efficiency).

Signal-portability check: reward substitution

To test whether the improvement originates at the signal layer rather than in a particular pipeline, we substitute $\hat{\varphi}$ for a manager’s original outcome reward, leaving its optimization otherwise unchanged (Table 2). We test whether this substitution lifts two reinforcement-style managers by $[TBD]$ and $[TBD]$ respectively. This is a portability check on one signal, not a separate method; the full table is a candidate for the appendix.

Ablation

Table 3 removes one component at a time and adds two alternative-signal controls. We read it as a mechanism-consistent sequence—credit fidelity (CCC) $\rightarrow$ mechanism and interference change (SR@20%, CTI) $\rightarrow$ task outcome (AVG)—rather than as isolated cells or a proven mediation. The pivotal test is A2: the identical ACA architecture, features, update rule, and compute budget, trained on a matched observational/outcome label instead of the interventional label. If the gain persists under A2, the contribution is not localized to interventional supervision, and the interpretation must be revised.

Mechanism and calibration

Figure 5 reuses the axes and statistical definitions of the baseline diagnostic (Figure 2) so that the two are directly comparable. Panel (a) tests whether ACA’s held-out credit-contribution correlation $[TBD]$ improves on the baseline $[TBD]$; the source gate for H2 is a Spearman correlation of at least 0.6. Panel (b) tests whether the superstition rate is

Method	S1	S2	S3	S4	S5	W-A	LME	LCM	AVG
No-Memory ReAct	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Full-History Stuffing	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Mem0	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A-MEM	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
AWM	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
ReasoningBank	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
SkeMex	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Memory-R1	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Mem- α	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
MemRL	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Reflexion	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
ReasoningBank+MaTTS	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
MERIT (ours)	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
RIT-Full reference	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>

Table 1: Main-results skeleton with 14 method rows. S1–S5 are ordered by planned stream heterogeneity; the identities of S2–S4 remain unspecified in the source. All result cells are unresolved.

Manager / reward	AVG	Δ original
Memory-R1 / original	<i>TBD</i>	–
Memory-R1 / $\hat{\phi}$	<i>TBD</i>	<i>TBD</i>
Mem- α / original	<i>TBD</i>	–
Mem- α / $\hat{\phi}$	<i>TBD</i>	<i>TBD</i>

Table 2: Reward-substitution skeleton. The table will test signal portability; it does not yet establish that any improvement originates at the signal layer.

Fig. 5 placeholder: Mechanism recovery

(a) x : predicted contribution $\hat{\phi}$; y : RIT contribution ϕ .
(b) x : task index t (100–500); y : SR@20%.
Compare ACA (MERIT) with the correlational baseline.
Axis ranges and statistical definitions must be identical to Fig. 2.

Figure 5: Planned mechanism-closure analysis comparing attribution fidelity and superstition-rate trajectories under the same definitions and axes as the baseline diagnostic.

flattened to *[TBD]*, a reduction of *[TBD]* relative to the baseline at $t = 500$; the H1 reading is the *slope* of this trajectory, compared between the full method and the no-governance ablation across the deployment checkpoints. Calibration is assessed with a held-out reliability diagram and expected calibration error *[TBD]*, per-domain calibration, and a split-half re-test, with particular attention to sign instability near $\varphi \approx 0$.

Boundary and scaling

We test whether the gain holds under a larger backbone (from *[TBD]* to *[TBD]*), with stronger baselines but an unchanged

superstition mechanism) and whether an attributor trained on the pilot environments transfers zero-shot, measured by the transfer correlation *[TBD]* (H3). For the boundary (H5), we plan three trend lines: gain versus bank size (the small-bank bin has planned gain *[TBD]*), gain versus stream heterogeneity (single-domain gain *[TBD]*), and gain versus injected redundancy. The redundancy trend has no registered result identifier and is reported as an EVIDENCE GAP, kept exploratory; the hypotheses are that leave-one-out underestimation grows with redundancy and that the group-intervention variant (A7) mitigates it.

Efficiency

Figure 4 plots cumulative success over the stream and, as an inset shown only after verification, a planned RIT-budget sweep over $p \in \{1, 2, 5, 10\}\%$; the inset is labeled a planned comparison rather than a measured curve. We test whether performance differences concentrate in the later, most-polluted portion of the stream—what share *[TBD]* of the average gain arises there—and whether the RIT budget saturates near the source default, placing MERIT at a Pareto-efficient knee (H4). Table 6 records relative token cost, success, latency, and memory overhead against the co-occurrence baseline and the intervention reference.

Case study

Figure 6 plans two qualitative timelines. In the success case, a useful multi-hop memory and a superstitious “retry the same query” memory are tracked over deployment; we test whether the latter’s inflated utility is assigned near-zero RIT credit and leads to eviction at task *[TBD]*, while the former’s scope is characterized correctly. In the failure case, two complementary memories each hold half of a procedure; we test whether individual leave-one-out is near zero for both so that governance nearly removes both, and whether the group-intervention fallback recovers their joint credit—an honest display of the AS3 boundary. We expect a share *[TBD]* of MERIT’s remaining failures to stem from the bank simply lacking relevant experience, which no credit signal

Variant	AVG	Δ AVG	SR@20% ($t = 500$)	CCC	CTI
Full MERIT	<i>TBD</i>	0	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A1: no recalibration	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A2: matched obs.-label control	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A3: no scope gating	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A4: no governance	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A5: no usage features	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A-judge: LLM self-judge	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A7: group / sampled Shapley	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>

Table 3: Ablation skeleton. The table tests credit fidelity, mechanism statistics, interference, and task outcomes without pre-supposing that any component is necessary. A2 is a capacity-matched control: the identical ACA architecture, features, update rule, and compute budget, with only the supervision label swapped from the interventional RIT label to a matched observational/outcome label.

Gate	Estimand	Comparator	Audit pop.	Decision rule	Failure action
G-C1	SR@20% slope over t ; memory-level CCC($\hat{U}, \hat{\Phi}$)	baseline banks	pilot audit pool	slope CI excludes 0 (rises) and CCC below the pilot bound (> 0.5 falsifies)	narrow C1 to heterogeneous streams
G-C2	event-level CCC($\hat{\varphi}, \hat{\varphi}$); memory-level CCC($\hat{\varphi}, \hat{\Phi}$)	correlational baseline	held-out audit split	event-level $\rho \geq 0.6$ with a CI rule	C2 unsupported
G-C3a	Full-vs-A4 SR@20% slope difference	A4 (no governance)	mechanism audit	slope-difference CI excludes 0 (Holm)	drop flattening claim
G-C3b	CTI with vs. without scope gating	A3 (no scope)	A/B mixed-stream	CTI-reduction CI excludes 0 (Holm)	drop interference claim
G-C3c	AVG lift from $\hat{\varphi}$ -reward substitution	original reward	reward-swap runs	lift CI excludes 0 (Holm)	report null portability
G-C4	AVG gain over the simultaneous max baseline	max-baseline contrast	main runs, macro-AVG	MERIT $>$ max-baseline, Holm-corrected	report parity
G-H4	Pareto knee on the budget-AVG curve	budget sweep	efficiency runs	knee exists per the marginal-gain rule	drop Pareto claim
G-H5	directional gain vs. bank size / heterogeneity / redundancy	within-axis	boundary runs	predicted sign holds (CI); redundancy is Evidence Gap	scope H5 to passing axes

Table 4: Atomic falsification gates for C1–C4 (with H4/H5). Gates G-C1 (superstition), G-C2 (ACA fidelity), G-C3a–c (governance / scope / reward portability), G-C4 (end-task superiority), G-H4 (efficiency), G-H5 (boundaries). This table is design metadata; it contains no measured results, and its estimands, comparators, and CI/threshold rules are locked before any run.

can repair.

Limitations and Conclusion

Limitations. MERIT inherits three scope boundaries. It depends on local replayability (AS2): fully non-replayable, single-shot environments cannot yield RIT labels, and world-model-simulated replay or off-policy evaluation would be needed as a substitute. It depends on bounded interaction (AS3): leave-one-out underestimates credit under high redundancy, and the group-intervention fallback adds cost. And its benefit has a boundary (H5): single-domain short streams and small banks are expected to yield little gain, so MERIT is intended for long-term deployment rather than short sessions. We further scope the contribution to credit assignment for existing memories and leave write-time quality out of scope; MERIT composes with methods that address it. A rebuttal-sensitive risk is that, if attribution scores can be influenced by externally supplied content, the audit becomes a new attack surface for memory-poisoning “credit launder-

ing,” which we flag for future defense. Whether ACA transfers across fundamentally different modalities beyond the tested text and agentic environments is not specified by our source and is left as an open question, excluded from C1–C4.

Conclusion. We argued that self-evolving agent memory is limited less by better heuristics than by a missing primitive: causal credit. Correlational bookkeeping mistakes retrieval for contribution and, we hypothesize, breeds superstitious memories that accumulate over deployment. MERIT is designed to supply counterfactual credit affordably by collecting small-budget interventional labels (RIT), amortizing them into an $O(1)$ -per-event score (ACA), and exposing that single score to two minimal consumers. The evaluation is planned rather than completed: it tests whether ACA recovers the audited signal in closed loop, whether its consumers flatten superstition and reduce cross-task interference, and whether the recovered signal improves average success at low overhead—each behind an explicit falsification gate. Two concrete directions follow: structure-aware interaction attri-

Method	Causal signal	O(1) online	Recalibration	Scope	Audit
Data Shapley	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Influence Functions	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
TracIn	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
ContextCite	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
ReasoningBank	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
SkeMex	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Memory-R1	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
MemRL	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
MERIT (ours)	yes	yes	yes	yes	yes

Table 5: Method-positioning skeleton. Every competitor cell remains TBD until its cited source is checked item by item; TBD must not be interpreted as lack of capability.

Configuration	Relative tokens	AVG	ACA ms/event	Extra VRAM	RIT overhead
Co-occurrence baseline	$1\times$	<i>TBD</i>	–	–	0
MERIT	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
A1 (static attributor)	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
RIT-Full reference	<i>TBD</i>	<i>TBD</i>	–	–	<i>TBD</i>

Table 6: Efficiency skeleton. All non-reference values remain unresolved; no Pareto ranking is assigned before verification.

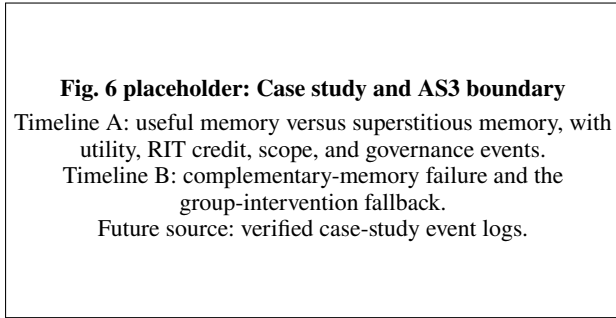


Figure 6: Planned qualitative analysis of a successful governance case and a complementary-memory failure that exposes the AS3 boundary.

bution to relax AS3, and replay substitutes to relax AS2.